Competing Poisson Flows: A Derivation Journey

Not a solution report — a record of how the answer was *earned*

Order-flow buy/sell races, derived from axioms, with every wrong turn kept in

The problem. Buys arrive as a Poisson process of rate λ; sells as an independent Poisson process of rate μ. Three questions: (a) probability the next event is a buy; (b) given the next is a buy, the time-until-arrival distribution; (c) probability that 3 buys arrive before 2 sells.

Why this document looks unusual. A clean solution would be three boxed formulas. That hides the part that actually transfers to new problems: the *reasoning discipline*. So this version keeps the naive first attempts, the missteps, the diagnosis of each, the fix, and — the real payload — the *generalization* each step unlocks. The boxed final answers are the least valuable thing here.

1 The methodological stance that drove everything

Before any computation, one commitment shaped the whole derivation: **first principles, no reverse-engineering, no approximation**. Concretely:

- Do not presuppose the exponential law $\mathrm{Exp}(\lambda+\mu)$ as "known." Derive it.
- Do not work backwards from the textbook answer $\frac{\lambda}{\lambda+\mu}$ and dress it up.
- When a small-order term looks negligible, do not wave it away — prove it vanishes.

This stance is expensive (it forces the long road through Kolmogorov's ODEs and an ε–style squeeze) but it is exactly what separates "I can recite the result" from "I can rebuild the result on a whiteboard under adversarial questioning."

2 Foundation: deriving the counting law from axioms

We take a Poisson process of rate r to be *defined* by three properties: **(P1)** independent increments; **(P2)** stationary increments; **(P3)** orderliness, $\mathbb{P}(N_h = 1) = rh + o(h)$ and $\mathbb{P}(N_h \geq 2) = o(h)$.

2.1 The empty- and one-event probabilities

Splitting $[0, t+h] = [0,t] \cup [t, t+h]$ and using independent increments, $P_0(t) := \mathbb{P}(N_t = 0)$ satisfies $P_0(t+h) = P_0(t)(1 - rh + o(h))$, hence

$$P_0'(t) = -r\,P_0(t),\ P_0(0) = 1 \implies \boxed{P_0(t) = e^{-rt}.}$$

The exponential is *produced* by the ODE. For one event, decomposing on where it falls,

$$P_1'(t) = -r\,P_1(t) + r\,P_0(t),\ P_1(0) = 0 \implies \boxed{P_1(t) = rt\,e^{-rt}.}$$

Generalize — this is Kolmogorov's forward equation

The recursion $P_n'(t) = -rP_n(t) + rP_{n-1}(t)$ is not special to Poisson. It is the *Kolmogorov forward (Fokker–Planck) equation* for a pure-birth chain. Replace the constant rate r by state-dependent rates and the *same machine* delivers birth–death processes, $M/M/c$ queues, and chemical reaction networks. Poisson is the simplest instance: rates constant. That is why this route feels heavy — it is a general-purpose engine being run on a toy.

2.2 A subtlety worth keeping: why "≥ 2 events" must be considered

When we later compute things in a small window, the tempting shortcut is: "the event I want is *1 buy, 0 sell*, so that's the whole story." It is not.

The first instinct

Model "first event is a buy in this cell" as exactly "$b = 1, s = 0$ in the cell," and stop.

Why that is incomplete

"First event is a buy" is a *strict superset* of "exactly 1 buy, 0 sell." The cell could contain ≥ 2 events with a buy in front (2 buys; 1 buy 1 sell with buy first; ...) — all of those still have a buy as the first event. Honestly accounting for "first is a buy" *forces* us to include the ≥ 2 mass, call it $\rho(\Delta)$. Silently dropping it assumes, without proof, that two arrivals can't share a cell — false for finite Δ.

The fix is a **squeeze**, not an approximation. We never compute $\rho(\Delta)$; we trap it:

$$0 \leq \rho(\Delta) \leq \mathbb{P}(\geq 2 \text{ in cell}) = 1 - e^{-\beta} - \beta e^{-\beta}, \qquad \beta := (\lambda + \mu)\Delta.$$

The bound came from monotonicity ($\{\geq 2 \text{ and buy first}\} \subseteq \{\geq 2\}$, drop the "buy first" constraint to get a count-only event we can compute exactly via the Poisson $k = 0, 1$ terms). Expanding $e^{-\beta} = 1 - \beta + \frac{\beta^2}{2} - \cdots$ exactly, $\mathbb{P}(\geq 2) = \frac{\beta^2}{2} + O(\beta^3) = O(\Delta^2)$, so $\rho(\Delta)/\Delta \to 0$. The dirty mass is killed *exactly* in the limit, by squeeze and the exact Poisson series — no hand-waving.

3 Philosophical detour: what does "first principles" even mean?

Mid-derivation a real question surfaced: *which* route is the true first-principles one — the ODE route above, or the short "min of two exponentials" route? The instinct was "the ODE route, because it presupposes less (it doesn't even assume the exponential)."

The instinct is backwards

Count the assumptions. The ODE route needs *four* axioms (independent + stationary increments + orderliness + a finite rate). The min route needs the exponential law, which itself follows from a *single* axiom — **memorylessness**, $\mathbb{P}(\tau > s + t) = \mathbb{P}(\tau > s)\mathbb{P}(\tau > t)$, whose only solution is $e^{-\lambda t}$ (Cauchy's functional equation). And the four Poisson axioms *imply* memorylessness of the gaps, while memorylessness does *not* imply independent increments or orderliness. So $\{\text{ODE axioms}\} \supsetneq \{\text{min-route axioms}\}$: the ODE route assumes strictly *more*.

Two confusions were untangled:

- **"Not presupposing" ≠ "fewer assumptions."** The ODE route does more *work* (it re-derives the exponential), but it carries a heavier set of premises in at the door. Doing more work is not the same as assuming less.
- **Logical depth ≠ historical order.** Historically the exponential/Poisson *distributions* predate the "Poisson process" as an axiomatized object by a century. That is not an accident: the memorylessness path to the exponential is genuinely shorter. Axiomatization is usually the *endpoint* of understanding, not its starting gun.

Generalize — the real value of each route

The ODE route's virtue is not "more fundamental" — it is *universal*. Kolmogorov's forward equation handles *every* continuous-time Markov chain; Poisson is one special case. The min/memorylessness route is the *sharp, specialized* tool for the exponential structure. Analogy: Gauss's pairing trick for $1 + \cdots + n$ versus the Euler–Maclaurin machine. Both correct; one is specialized and fast, one is general and heavy. Neither is "more true."

4 Part (a): deriving p from zero, refusing to reverse-engineer

The first move

Write $\mathbb{P}(b = 1) = \mathbb{P}(b = 1, s = 0) + \mathbb{P}(b = 1, s = 1)$, then drop the second term as small.

Two things wrong

(i) The "$+\mathbb{P}(b = 1, s = 1)$" term is not merely small — under orderliness it is *exactly* zero (simultaneous buy and sell in an infinitesimal window has probability $O(\Delta t^2)$, vanishing relative to the events we keep). The instinct to drop it was right; the *reason* was wrong. (ii) More importantly, $\mathbb{P}(b = 1, s = 0 \mid \Delta t)$ is a *window count*, scaling like Δt and tending to 0 — it is not the ratio p we want. The object we actually want is a *conditional* probability.

The reframe: $p = \mathbb{P}(\text{it's a buy} \mid \text{one event occurred})$. Let $A = \{b = 1, s = 0\}$ and $D = \{\text{exactly one event}\}$. Since one buy *is* one event, $A \subseteq D$, so $A \cap D = A$; and D splits into the mutually exclusive "one buy" / "one sell." By the **definition of conditional probability** plus additivity over the disjoint denominator:

$$p = \mathbb{P}(A \mid D) = \frac{\mathbb{P}(A \cap D)}{\mathbb{P}(D)} = \frac{\mathbb{P}(b = 1, s = 0)}{\mathbb{P}(b = 1, s = 0) + \mathbb{P}(b = 0, s = 1)}.$$

This is the definition, not Bayes

Bayes reverses a conditioning direction using prior × likelihood / evidence. Here nothing is reversed: we use $\mathbb{P}(A \mid D) = \mathbb{P}(A \cap D)/\mathbb{P}(D)$ raw, with $A \subseteq D$ collapsing the numerator and mutual exclusivity expanding the denominator. No prior, no likelihood, no Bayes.

Plug in the *exact* Poisson probabilities (independence ⇒ multiply):

$$\mathbb{P}(b = 1, s = 0 \mid \Delta t) = \lambda \Delta t\, e^{-(\lambda+\mu)\Delta t}, \qquad \mathbb{P}(b = 0, s = 1 \mid \Delta t) = \mu \Delta t\, e^{-(\lambda+\mu)\Delta t}.$$

Limit by cancellation — not L'Hôpital

The instinct to "apply L'Hôpital to the 0/0" was unnecessary. The common factors Δt and $e^{-(\lambda+\mu)\Delta t}$ sit in numerator *and* denominator and cancel *before* any limit:

$$p = \lim_{\Delta t \to 0} \frac{\lambda \cancel{\Delta t}\, \cancel{e^{-(\lambda+\mu)\Delta t}}}{(\lambda+\mu)\, \cancel{\Delta t}\, \cancel{e^{-(\lambda+\mu)\Delta t}}} = \boxed{\frac{\lambda}{\lambda+\mu}.}$$

After cancellation nothing depends on Δt, so there is no indeterminate form to resolve. This is the dividend of having kept the exponential factor *exact* back in the squeeze step: it now cancels cleanly instead of needing to be approximated.

Generalize — k competing streams

Nothing used "two." With k independent Poisson streams of rates $\lambda_1, \ldots, \lambda_k$, the identical argument gives the probability that the next event is of type i:

$$p_i = \frac{\lambda_i}{\sum_{j=1}^{k} \lambda_j}.$$

The conditional-probability-definition scaffold (numerator = target stream, denominator = sum over all streams, common $\Delta t\, e^{-(\sum \lambda)\Delta t}$ cancels) is stream-count agnostic. In an order book this is "which venue / which order type fires next."

5 Part (b): survival $\times$ hazard, and the factorization that is the real prize

The first move

Write the event "first event is a buy at T" as $\mathbb{P}(b=0, s=0,\, t<T) + \mathbb{P}(b=1,\, t=T)$, and try to evaluate each piece by integrating over t.

Three errors, each instructive

(i) **"+" should be "×."** "Survive $[0,T)$" *and* "a buy fires at T" is an *intersection* of independent events — probabilities multiply, they do not add. "+" is for mutually exclusive alternatives, of which there are none here.
(ii) **Survival is a probability, not an integral.** "$[0,T)$ empty" is a *single* event with value $e^{-(\lambda+\mu)T}$. Writing it as $\int_0^T e^{-(\lambda+\mu)t} dt$ produces a quantity carrying units of time — a tell-tale sign it cannot be a probability.
(iii) **The buy term dropped μ.** Using $\lambda e^{-\lambda t}$ for "buy fires" forgets that the sell must *also* not have fired ($e^{-\mu t}$). The giveaway: a first-arrival time scale *must* depend on $\lambda+\mu$, never on λ alone.

The correct object is built multiplicatively, survival $\times$ hazard, and is a *density*:

$$f_{\text{buy}}(T)\, dT = \underbrace{e^{-(\lambda+\mu)T}}_{\text{survival (a probability)}} \cdot \underbrace{\lambda\, dT}_{\text{hazard} \times dT} \implies \boxed{f_{\text{buy}}(T) = \lambda e^{-(\lambda+\mu)T}.}$$

5.1 The measure-zero subtlety, resolved cleanly

A genuine worry arose: if $\mathbb{P}(\tau = T) = 0$ exactly, doesn't the density limit become meaningless?

Two different zeros

$\mathbb{P}(\tau = T) = 0$ is the *result* of a completed limit (a single point in continuous time; exactly zero by countable additivity). The Δ inside the density limit is the *process*: it is strictly positive throughout, and we *cancel* it against the Δ in the numerator *before* sending it to 0. A derivative never plugs in $\Delta = 0$; it asks where the ratio *tends*. So the limit is meaningful precisely because 0 is the destination, never a value substituted in. The felt contradiction came from pushing "the result's zero" back into "the process." They are not the same zero.

5.2 Normalize, then read off independence

$\mathbb{P}(M = \text{buy}) = \int_0^\infty \lambda e^{-(\lambda+\mu)T} dT = \frac{\lambda}{\lambda+\mu}$ (same as (a), now by integration). By the definition of conditional density,

$$f_\tau(T \mid M = \text{buy}) = \frac{\lambda e^{-(\lambda+\mu)T}}{\lambda/(\lambda+\mu)} = (\lambda+\mu)e^{-(\lambda+\mu)T} \implies \boxed{\tau \mid \{M = \text{buy}\} \sim \text{Exp}(\lambda+\mu).}$$

Now the prize. The joint density *factors*:

$$\underbrace{\lambda e^{-(\lambda+\mu)T}}_{f_{\text{buy}}(T)} = \underbrace{(\lambda+\mu)e^{-(\lambda+\mu)T}}_{f_\tau(T)} \times \underbrace{\frac{\lambda}{\lambda+\mu}}_{\mathbb{P}(M=\text{buy})}.$$

A joint density that factors into (function of T) $\times$ (function of type) *is the definition* of independence. So **the first-arrival time τ is independent of the first-event type M.** Note the direction we earned this in: we computed the joint object and *read off* independence — we did not assume independence and retrofit it.

Generalize — the competing-risks first-passage template

Survival $\times$ hazard is the universal skeleton for any "survive until something fires" problem. For k streams, the density of "stream i fires first, at t" is $\left(\prod_j e^{-\lambda_j t}\right)\lambda_i = \lambda_i e^{-(\sum_j \lambda_j)t}$, which factors as $\left(\sum_j \lambda_j\right)e^{-(\sum\lambda)t} \cdot \frac{\lambda_i}{\sum\lambda}$ — so first-time $\perp$ first-type holds for *any* number of competing exponential streams. This is the backbone of reliability theory, queueing, and any "which risk realizes, and when" question.

Whiteboard version (use this live). $\tau = \min(\tau_b, \tau_s)$, $\mathbb{P}(\tau > t) = e^{-\lambda t}e^{-\mu t} \Rightarrow \tau \sim \text{Exp}(\lambda+\mu)$; and $\mathbb{P}(\tau_b < \tau_s) = \int_0^\infty \lambda e^{-\lambda s}e^{-\mu s} ds = \frac{\lambda}{\lambda+\mu}$. Three lines. Keep the factorization argument in reserve for "why are time and type independent?"

6 Part (c): the counting race, enumerated honestly

By memorylessness each arrival is independently a buy ($p = \frac{\lambda}{\lambda+\mu}$) or a sell ($q = 1-p$); by part (b) timing is irrelevant. The race is a string of i.i.d. labels.

The first move (good instinct, imprecise words)

"Five events, the last one a buy, the middle four in any order."

Two corrections

(i) **"Any order" is too loose.** The order of the pre-final events is free, but their *type composition* is pinned: before the deciding buy there must be exactly 2 buys, so that the final buy is the 3rd. (ii) **The race can end early.** If the 2nd sell lands first, the seller wins and there is no 5th event. So "buyer wins" is not one fixed-length path; it splits by how many sells precede the final buy — and that count is at most 1.

The user then sharpened it correctly: *the sequence ends on the 3rd buy, and before it the sells number 0 or 1* ("before each sell there must already be a buy" was the intuition; the precise statement is "pre-final sells ≤ 1"). Two mutually exclusive cases:

$$\underbrace{p^3}_{\text{0 sells: } bbb} + \underbrace{\binom{3}{1}p^3q}_{\text{1 sell among the 3 pre-final slots}} = p^3(1+3q).$$

Why $\binom{3}{1}$, not $\binom{4}{1}$, and no extra 2!

The final buy is *pinned* at the end and does not get permuted; only the 3 events before it do, so the sell chooses 1 of 3 slots. Allowing 4 slots would illegally count sequences with the sell *after* the deciding buy. And we count *type* sequences, so the two buys are not separately labelled — no factor of 2! for their internal order.

Substituting $p = \frac{\lambda}{\lambda+\mu}$, $q = \frac{\mu}{\lambda+\mu}$:

$$\boxed{\mathbb{P}(\text{3 buys before 2 sells}) = p^3(1+3q) = \frac{\lambda^3(\lambda+4\mu)}{(\lambda+\mu)^4}.}$$

Generalize — n buys before m sells (negative-binomial race)

The race is decided no later than the $(n+m-1)$-th event: by then either $\ge n$ buys or $\ge m$ sells have occurred. "n buys before m sells" = "at least n buys among the first $n+m-1$ labels." Hence

$$\mathbb{P}(n \text{ buys before } m \text{ sells}) = \sum_{k=n}^{n+m-1} \binom{n+m-1}{k} p^k q^{n+m-1-k}.$$

Check: $n=3, m=2 \Rightarrow \sum_{k=3}^{4}\binom{4}{k}p^kq^{4-k} = \binom{4}{3}p^3q + \binom{4}{4}p^4 = p^3(4q+p) = p^3(1+3q)$ since $p+q=1$. *Equivalent view:* this is exactly a banded Bernoulli/negative-binomial "who reaches their threshold first" race — the same structure as Pólya urns and best-of-N series.

7 The independence audit — and where the whole thing breaks

The compact answers rest on a chain of independence, worth naming explicitly because it is also the *fault line*:

- two processes independent (given) $\Rightarrow$ the split $p = \frac{\lambda}{\lambda+\mu}$;
- memorylessness $\Rightarrow$ each arrival's type is i.i.d. (the race restarts fresh each time);
- time $\perp$ type (part (b)) $\Rightarrow$ discard the time axis, look only at labels.

Generalize — break the assumption on purpose

Replace the Poisson buys with a **Hawkes (self-exciting)** process, where each buy raises the intensity of future buys ($\lambda_t = \lambda_0 + \sum_{t_i<t} \phi(t-t_i)$). Then p is no longer constant, labels are no longer i.i.d., and time and type are no longer independent — so the factorization in (b) and the i.i.d. collapse in (c) both fail. The clustering of order flow that HFT desks actually trade against lives *precisely* in this broken regime. Knowing exactly which assumption fails (and therefore which formula to stop trusting) is the point of having derived them from first principles rather than memorizing them.

8 What actually transfers

The three boxed answers — $\frac{\lambda}{\lambda+\mu}$, $\mathrm{Exp}(\lambda+\mu)$, $p^3(1+3q)$ — are the disposable part. The transferable assets are the reasoning moves and the reflexes for catching your own errors:

- **Conditional-probability definition** vs Bayes — know which one you are using and why.
- **Survival $\times$ hazard** (multiply, not add) for any first-passage density; the factorization that exposes time $\perp$ type.
- **Probability vs density** — single points have probability 0; the density is the live object; the limit is meaningful because Δ stays positive in the process.
- **Squeeze, not approximate** — trap the $O(\Delta^2)$ junk and prove it dies.
- **Cancellation, not L'Hôpital** — keep factors exact so they cancel cleanly.
- **Pin the terminal event** in a race; permute only what precedes it; count *type* sequences.
- **Know which assumption you are leaning on**, so you know exactly when (e.g. under Hawkes) the answer stops being valid.